AI-Based Universal JEE Question Generation System

Multimodal AI Pipeline for Automated JEE/NEET Variant Generation

Goal: Transform a single expert-authored JEE question image into unlimited mathematically-valid question variants using multimodal AI, symbolic validation, and deterministic rendering.

INPUT IMAGE	GPT-4 VISION	JSON ENGINE	QWEN2-VL	FINAL OUTPUT
JEE Question Upload	Semantic Extraction	Constraint Validation	Layout-aware Editing	Generated Variants

1. System Vision & Architecture

The proposed architecture enables scalable generation of visually consistent and mathematically-valid JEE/NEET question variants. The system combines semantic extraction, symbolic mathematics, multimodal layout understanding, and deterministic rendering to create high-quality educational content automatically.

Core Capability	Description
Semantic Understanding	GPT-4 Vision extracts formulas and constraints
Constraint Validation	SymPy validates generated parameters
Layout-aware Editing	Qwen2-VL maps image regions precisely
Deterministic Rendering	Pillow + SVG generate clean readable output

2. Unified AI Pipeline

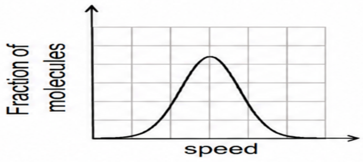
Layer	Technology	Purpose
Input Layer	Question Image	Original JEE/NEET question
Semantic Layer	GPT-4 Vision	JSON extraction
Validation Layer	SymPy	Constraint verification
Layout Layer	Qwen2-VL	Region-level planning
Rendering Layer	Pillow + SVG	Diagram generation
Output Layer	PNG/PDF Export	Final generated variants

3. Generated Question Variants

The examples below demonstrate visually consistent, AI-generated JEE question variants created from semantic JSON templates and layout-aware rendering.

Example 1 — Statistical Mechanics Question Variants

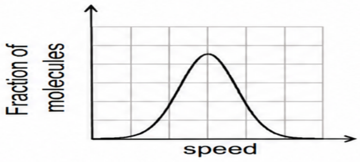
Q.1 If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average, and the root mean square speeds, respectively, is



(A) 1 : 1 : 1
(B) 1 : 1 : 1.224
(C) 1 : 1.128 : 1.224
(D) 1 : 1.128 : 1

Answer: (C)

Q.1 If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average, and the root mean square speeds, respectively, is



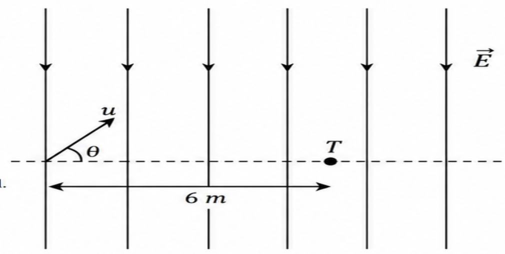
(A) 1 : 1 : 1
(B) 1 : 1 : 1.250
(C) 1 : 1.150 : 1.250
(D) 1 : 1.150 : 1

Answer: (C)

Example 2 — Electrodynamics Question Variants

1

A uniform electric field, $\vec{E} = -500\sqrt{3}\,\hat{y}$ NC^{-1} is applied in a region. A positively charged particle having $\frac{q}{m} = 2 \times 10^{10}$ Ckg^{-1} is projected with an initial speed $3\sqrt{5} \times 10^{6}$ ms^{-1}. The particle is aimed to hit a target T, which is 6 m away from its entry point into the field as shown in the figure.



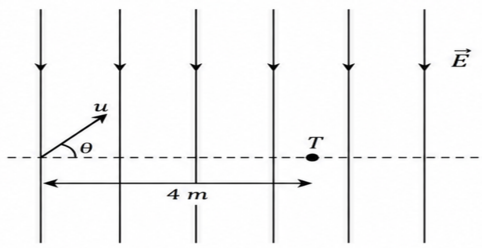
Then, the correct statements are

- A The particle will hit T if projected at an angle 45° from the horizontal.
- B The particle will hit T if projected either at an angle 30° or 60° from the horizontal.
- C The time taken by the particle to hit T could be $\sqrt{\frac{3}{5}}$ μs as well as $\sqrt{\frac{9}{5}}$ μs.
- D The time taken by the particle to hit T is $\sqrt{2}$ μs.

Correct Answer: B, C

2

A uniform electric field, $\vec{E} = -300\sqrt{3}\,\hat{y}$ NC^{-1} is applied in a region. A positively charged particle having $\frac{q}{m} = 8 \times 10^{9}$ Ckg^{-1} is projected with an initial speed $4\sqrt{2} \times 10^{6}$ ms^{-1}. The particle is aimed to hit a target T, which is 4 m away from its entry point into the field as shown in the figure.



Then, the correct statements are

- A The particle will hit T if projected at an angle 45° from the horizontal.
- B The particle will hit T if projected either at an angle 30° or 60° from the horizontal.
- C The time taken by the particle to hit T could be $\sqrt{\frac{2}{3}}$ μs as well as $\sqrt{2}$ μs.
- D Only one trajectory exists for the particle to hit T.

Correct Answer: B, C

4. JSON-guided Variant Generation Workflow

Input	Transformation	Output
Question Image	GPT-4 Vision Extraction	Semantic JSON
Semantic JSON	Constraint Validation	Validated Parameters
Validated JSON	Qwen2-VL Region Mapping	Edit Instructions
Edit Instructions	Pillow + SVG Rendering	Generated Variant

The complete pipeline enables scalable generation of visually rich and mathematically-valid educational content suitable for adaptive learning systems, personalized practice platforms, and AI-assisted EdTech workflows.